

Muhammad Yousif

Full-Stack Developer · Computer Systems Engineer · Embedded Systems

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Professional Summary

Detail-oriented **Computer Systems Engineering** student with extensive hands-on experience in full-stack application architecture, embedded software design, and relational database systems. Built a production-ready full-stack platform using **Node.js**, **Express**, **Prisma ORM**, and **Flutter**, integrating robust authentication mechanisms and secure API routing. Proven ability to construct specialized engineering solutions, ranging from embedded **C/C++ logic on ESP32 microcontrollers** with real-time hardware control to automated **desktop interfaces with Java Swing** and MySQL. Solid foundation in software engineering patterns, algorithmic optimizations, and database normalization with a focus on writing clean, secure, and production-grade code.

Technical Skills

Languages: Python, Java, C, C++, SQL, JavaScript (Node.js), Dart, Bash

Full-Stack & Frameworks: React JS, Flutter, Express, Prisma ORM, Java Swing

Databases & Architecture: MySQL, PostgreSQL, JWT Authentication, API Routing, Left/Right Outer Joins

Embedded & Robotics: ESP32, Raspberry Pi, FreeRTOS, Microcontroller Peripherals, Embedded C

Developer Tools & Security: Git, GitHub Actions, Docker, Helmet.js, Express Rate Limit, Firebase Cloud Messaging

Engineering Projects

Hostel Mess Management System — Full-Stack Platform

Node.js · Express · Prisma · Flutter · Dart · MySQL

github.com/Yousif-Sahito/hostel-mess-system

- **Architected** a comprehensive full-stack platform featuring a cross-platform Flutter frontend and a Node.js + Express backend.
- **Implemented** strict role-based access control (RBAC) via JWT tokens with a 7-day expiration and automated token caching to optimize network performance.
- **Secured** backend layers utilizing Helmet.js for HTTP header configuration and global rate limiting (1000 requests per 15 minutes), raising the overall security audit score to 8.6/10.
- **Engineered** real-time push notification pipelines via Firebase Cloud Messaging (FCM) with automated token updating routines upon app initialization.
- **Optimized** database transactions using Prisma ORM over MySQL, effectively blocking standard SQL injection vectors while coordinating data synchronization across 8+ system tables.
- **Refactored** client-side communications using interceptors to inject operational headers, fixing critical authentication path drops and routing misalignments.

Airline Management System — Desktop Application

Java · Java Swing · MySQL · JDBC · OOD

github.com/Yousif-Sahito/airline-management

- **Developed** a robust desktop-based enterprise automation platform utilizing Java Swing to streamline crucial administrative workflows.

- **Engineered** complex data-driven modules managing comprehensive flight bookings, ticket generation, records tracking, and instantaneous cancellations.
- **Integrated** a normalized MySQL database architecture linked via JDBC handlers, utilizing optimized standard Left and Right Outer Joins to cross-reference customer profiles with booking histories.
- **Structured** user interface choice boxes explicitly to render as empty prior to initial database handshake actions to preserve data processing integrity.

AI-Linked Color Detection Vehicle — Embedded Systems *Embedded C · C++ · ESP32 · Microcontrollers · Robotics*

- **Engineered** an autonomous embedded vehicle system driven by an ESP32 microcontroller programmatically orchestrated via Embedded C.
- **Interface-linked** a physical camera peripheral module directly onto the vehicle chassis to capture and transmit environmental video data streams.
- **Authored** local peripheral control routines executing real-time color categorization, triggering direct audio-voice announcements based on optical data inputs.

Experience

Full-Stack Developer Intern · Tech-Solutions Engineering (Remote)

June 2025 – Present

- **Maintained**, optimized, and tested full-stack code pipelines leveraging Node.js and React JS views to bolster platform scalability.
- **Implemented** secure input validation algorithms on public API endpoints, drastically mitigating data corruption patterns and error leakages.
- **Refactored** legacy relational queries, increasing backend data fetching speeds across multi-table schema landscapes.

Open Source & Published Tools

- **Hostel Mess App Backend** – Complete production-ready REST API repository built with Express and Prisma.
- **Airline System Core** – Open-source Java desktop client implementing swing-based dashboards.
- **ESP32 Color Assist** – Microcontroller firmware scripts written in C++ for edge color calculation.

Additional Technical Work

Discrete Mathematics Toolkit: Programmed localized mathematical scripts focusing on set theories and combinatorial evaluations to clarify academic processing pipelines.

Amazon Homepage Clone: Created a fully responsive webpage mockup utilizing modern HTML5 and CSS3 layout principles to refine responsive structural styling practices.

Education

B.E. Computer Systems Engineering

Expected 2028

Sukkur IBA University · Sukkur, Pakistan

Relevant Coursework: Data Structures and Algorithms, Database Management Systems, Operating Systems, Computer Communication Networks, Discrete Mathematics.